

QUESTIONS

1. The electric company charges according to the following rate schedule:

9 cents per kilowatt-hour (kWh) for the first 300 kWh
8 cents per kWh for the next 300 kWh (up to 600 kWh)
6 cents per kWh for the next 400 kWh (up to 1000 kWh)
5 cents per kWh for all electricity used over 1,000 kWh

Assume that maximum usage can be 2,000 kWh.

a) Which test inputs should be selected using equivalence class partitioning method? Construct a table showing equivalence partitions, the selected inputs and the expected outputs.

b) Which test inputs should be selected using boundary-value analysis method? Construct a table showing equivalence partitions, the selected inputs and the expected outputs.

2. Generate a decision table for an application that will determine whether the given year is a leap year or not. Below is the algorithm.

If year is divisible by 400 **then** is_leap_year
Else if year is divisible by 100 **then** not_leap_year
Else if year is divisible by 4 **then** is_leap_year
Else not_leap_year

3. Generate JUnit test class(es) for the following test case to test a method named Util.isPrime that takes an integer input and return a Boolean value. Note that you only read to write the test class not the Util.isPrime method.

Test case #	Input	Expected Output
1	2	True
2	4	False
3	0	False
4	5	True
5	-1	NumberFormatException

4. Generate test cases for the following use case:

Use case name: Session	ID: 1	Primary actor: Customer
Normal flow of events: 1. The user inserts the ATM card into the system. 2. The system starts the session and asks the user to insert the PIN. 3. The user enters his/her PIN number. 4. The system confirms the user and returns the transactions menu. 5. The user chooses the withdrawal option and enters the amount. 6. The system checks the account information 7. The requested money is given to the user and dropped from the account. The user is asked to do any other transactions 8. The user ends the session 9. The card is ejected		
Alternate/exceptional flows: 4A1: The user enters incorrect PIN to confirm his/her identity. • The system returns the user a warning about the incorrect PIN and asks to enter the correct one. • The user enters the correct PIN. • The use case continues. 4A2: The user enters incorrect PIN for 3 times. • The system returns the information about the situation. • The card is retained by the ATM and the session ends. 5A1: The user chooses to perform deposit operation. • The system asks the user to insert the cash. • The system checks the amount that is inserted. • The system asks the user to confirm the operation. • The amount is added to the account and the use case continues. 5A2: The user chooses to perform transfer operation. • The system asks the user to give the information about the account that the money will be transferred. • The user enters the account information. • The system asks for the amount to be transferred. • The user enters the amount. • The system checks the sender account's availability to send the requested amount. • The system transfers the money and drops the amount. • The use case continues. 5A3: The user chooses to perform inquiry. • The system returns the information about the account. • The use case continues		